

ANKA OS

Project Status Report — Update

What's Done, What's Left — Full Codebase Audit (v2.0)

Internal — Development Team

Frontend: anka-diversify-os (Next.js 16 / React 19) | Backend: anka-os-backend (Express 5 / Prisma / PostgreSQL)

Compiled 5 August 2026 — based on direct source inspection, HEAD commits 3 August 2026

Supersedes v1.0 (8 July 2026, based on HEAD commits 15 June 2026)

1. What Changed Since v1.0

Headline: the project is no longer dormant. v1.0 was compiled on the assumption that both repos had gone quiet after 15 June 2026 (23 days idle at the time). That has since reversed — active development resumed on **26 July 2026** and ran through **3 August 2026** (frontend +11 commits to 99 total, backend +12 commits to 70 total). This update re-audits the codebase against that new work.

The two pre-launch security blockers from v1.0 are both fixed

v1.0 finding	Status now
JWT auth middleware written but applied to zero routes — API reachable via any <code>X-User-ID</code> header	FIXED Commit 148fc39 (26 Jul) wraps <code>/api/ai</code> , <code>/api/admin</code> , <code>/api/projects</code> , <code>/api/notifications</code> , <code>/api/admin/rules</code> , <code>/api/admin/departments</code> in <code>authenticateToken</code> at the router-mount level in <code>src/index.ts</code> . Public-by-design routes (login, signup, invite validate/accept) apply it selectively per-route instead.
<code>ENCRYPTION_KEY</code> missing/short silently falls back to a hardcoded in-source key	FIXED Same commit adds <code>src/config/env.ts</code> , which calls <code>required("JWT_SECRET")</code> and <code>required("ENCRYPTION_KEY", 32)</code> and is imported first-line in <code>src/index.ts</code> — missing/short values now crash the process at startup instead of degrading silently. <code>encryption.ts</code> no longer contains a fallback key.

One of the two mock-data pages is fixed; the other is now honestly labeled

v1.0 finding	Status now
<code>/development</code> — every widget sourced from <code>lib/mock-data.ts</code> , presented as live	FIXED Commit 83706bd (26 Jul): now fetches real projects/tasks/sprints/commits via <code>projectApi</code> . The one piece with no real backing yet (historical sprint-velocity trend) is kept, but now explicitly labeled "Demo data" in the UI rather than presented as real.
<code>/product-modeling</code> — four fictional products hardcoded inline, no real backing	STILL MOCK, NOW LABELED Same commit adds a banner disclosing the content is sample data — there is still no backend model for product modeling, so the underlying gap is unresolved, but it no longer misrepresents itself as live.

A major new feature shipped that v1.0 never saw: the Phased Workflow

Commit 511463f (26 Jul, backend) plus a matching frontend push add a full **requirements** → **documentation** → **architecture** → **implementation** → **testing** → **review** pipeline, entirely new:

- **4 new Prisma models** — `ProjectPhaseState`, `PhaseArtifact`, `PhaseApproval`, `WorkflowRun` — all with real `onDelete: Cascade` relations to `Project` (notably *not* repeating the orphaning bug called out in v1.0 for chat/activity rows).

- **New route group** `/api/projects/:projectId/phases/*` (in `phase-routes.ts`, not yet in `CLAUDE.md`): list phase states, list/create artifacts, approval history, workflow runs, and per-phase `start` / `run` (AI drafts a proposal) / `request-approval` / `approve` / `request-changes` actions.
- **Frontend:** `components/project/phase-stepper.tsx`, an Architecture Phase view, brief-input + copy/download-PDF + revise-after-approval for phase docs, and a batch task runner with dependency ordering and mid-task Q&A.
- **Backend AI:** the coding agent gained task detail / architecture / memory / clarification awareness, and `ai-service.ts` nearly doubled — 1,378 → **2,373 lines** — across several same-day "implement AiService with OpenAI agentic tool support" commits (26–30 Jul), which read as iterative rewrites of the same feature rather than distinct additions.

This is a genuinely new subsystem, not a patch — it belongs in the main inventory (§3–§6 below), not just this changelog.

Unchanged from v1.0 — still open

Item	Status
No automated tests, no CI in either repo	UNCHANGED
<code>GET /api/projects/config/s3</code> debug endpoint (AWS config exposure)	PARTIAL route still exists and is not admin-gated, but is no longer reachable with zero identity — it now sits behind the global <code>authenticateToken</code> on <code>/api/projects</code> like every other route on that router, since fix #1 above applies at the mount point, not per-route.
Rules engine only executes 2 of the rule types it lets you create (<code>sprint_auto_close</code> , <code>overdue_escalation</code>)	UNCHANGED
Two parallel S3 implementations (<code>s3.service.ts</code> dead, reads wrong env var <code>AWS_S3_BUCKET_NAME</code>)	UNCHANGED — confirmed zero importers, still ships
Duplicate terminal implementations (real WebSocket one vs. simulated <code>/development/terminal</code>)	UNCHANGED
Orphaned frontend files: <code>app/api/ai/chat/route.ts</code> , <code>lib/enhanced-ai-service.ts</code> , <code>lib/context-manager.ts</code> , <code>lib/github-service.ts</code>	UNCHANGED
<code>lib/project-ai-service.ts</code> — literal 0-byte file	UNCHANGED — confirmed still 0 bytes
Duplicate signup components (<code>signup-form.tsx</code> / <code>signup-form-with-role.tsx</code>)	UNCHANGED
Duplicate AI chat route <code>/projects/[projectId]/ai</code> alongside the in-tab AI Assistant	UNCHANGED — page still present
<code>ProjectChatMessage</code> / <code>ProjectActivity</code> have no real Prisma relation to <code>Project</code> — orphan on delete	UNCHANGED — worth noting the team clearly knows how to do this correctly now, since all 4 new phase-workflow models use proper cascading relations

Internal docs (CLAUDE.md) understate the real route count

WORSE — the gap has grown; the new
/phases/* group (9 endpoints) isn't
documented either

2. Updated One-Paragraph Summary

Anka OS remains a substantial product, and it is **active again** after a six-week pause. The 26 Jul – 3 Aug push did two things at once: it closed the two items v1.0 flagged as pre-launch blockers (unauthenticated API, insecure encryption fallback) and one of its two fake-data pages, *and* it shipped an entirely new phased-workflow subsystem (requirements → documentation → architecture → implementation → testing → review) with its own AI-driven approval flow. The remaining gaps are almost entirely the same ones v1.0 named — no tests, a decorative rules engine, a handful of confirmed-dead files, duplicate terminal/signup/AI-chat implementations — none of which this push touched. The one new observation is a documentation debt now compounding on itself: CLAUDE.md and the internal route map were already stale in v1.0, and the new phase-workflow routes and models widen that gap further.

3. Data Model — Delta

23 → **27** Prisma models. All four additions belong to the phased workflow (see §1); everything else in v1.0's data-model inventory (§3 of v1.0) is unchanged, including the two flagged **RISK** items (`ProjectFile.size` as String; `ProjectChatMessage / ProjectActivity` with no cascade).

New model	Key fields	Notes
ProjectPhaseState	phase, status, approvedById?, approvedAt?	<code>@@unique([projectId, phase])</code> ; cascades on project delete
PhaseArtifact	phase, type, title, content (Text), version, approved	relates to <code>ProjectDecision[]</code> ; cascades on project delete
PhaseApproval	phase, approvedById, decision, comments?	append-only approval log; cascades on project delete
WorkflowRun	triggerType, currentPhase, status, modelUsage (Json), costUSD	tracks AI cost per phase run; cascades on project delete

4. Backend API — Delta

~50 → **~59** real endpoints. New group, mounted under each project and layered behind the now-global `authenticateToken`:

Method & Path	Purpose
GET <code>/api/projects/:id/phases</code>	List phase states for a project
GET <code>/api/projects/:id/phases/approvals?phase=</code>	Approval history for a phase
GET/POST <code>/api/projects/:id/phases/artifacts</code>	List / create phase artifacts (docs, diagrams, decision logs)

GET /api/projects/:id/phases/runs	Workflow run history (incl. AI cost/model usage)
POST /api/projects/:id/phases/:phase/run	AI drafts a phase proposal, logs a WorkflowRun
POST /api/projects/:id/phases/:phase/start	Mark a phase in-progress
POST /api/projects/:id/phases/:phase/request-approval	Move phase to awaiting_approval
POST /api/projects/:id/phases/:phase/approve	Approve a phase, records PhaseApproval
POST /api/projects/:id/phases/:phase/request-changes	Send phase back with feedback

All are new/undocumented relative to CLAUDE.md, same as the majority of v1.0's route inventory.

5. Backend Services — Delta

Service	What changed
middleware/auth.ts	FIXED now actually wired in at <code>src/index.ts</code> router-mount level for every non-public route group.
utils/encryption.ts	FIXED insecure fallback key removed; key is sourced from the now-required <code>ENCRYPTION_KEY</code> .
config/env.ts	NEW fail-fast startup validation for <code>JWT_SECRET</code> and <code>ENCRYPTION_KEY</code> (min length 32), imported before any module that signs/encrypts.
ai-service.ts	1,378 → 2,373 lines. Still gpt-4o (chat/agent) + gpt-4o-mini (cost-estimate paths) — CLAUDE.md's "gpt-4" claim for the coding agent remains inaccurate, and now also undersells the model actually used for phase-drafting calls. Gained phase-drafting, task-detail, architecture, memory, and clarification capabilities for the new workflow.
s3.service.ts	UNCHANGED — still reads <code>AWS_S3_BUCKET_NAME</code> , still has zero importers, confirmed dead.
rule-engine.ts	UNCHANGED — still exactly 2 executable rule types.
github.service.ts	Gained empty-repo handling: syncs gracefully against a repo with no commits yet, and initializes empty repos via the Contents API when the agent pushes changes.

6. Frontend Pages — Delta

Route / Component	Data	What changed
/development	FIXED	Now backend-wired for projects/tasks/sprints/commits; only the velocity trend remains placeholder, explicitly labeled.

/product-modeling	LABELED MOCK	Still sample content (no backend model exists), now disclosed via an on-page banner instead of presented as real.
Phase Stepper + Architecture Phase view	NEW	<code>components/project/phase-stepper.tsx</code> — drives the requirements→...→review UI; brief input, copy/download-PDF, and revise-after-approval for phase docs.
Batch task runner	NEW	Dependency-ordered batch execution of tasks with mid-task Q&A.
ProjectAIAssistant	NEW	<code>components/ai/project-ai-assistant.tsx</code> — multi-phase agent capabilities, loading/self-healing status reporting; appears to have gone through several same-week rewrites (4 commits, 29–30 Jul, similar messages) rather than one linear build.
/development/terminal (fake)	UNCHANGED	Still a fully simulated standalone page, coexisting with the real WebSocket terminal in the IDE tab.
/projects/[projectId]/ai (duplicate)	UNCHANGED	Still present alongside the in-tab AI Assistant.
signup-form.tsx / signup-form-with-role.tsx	UNCHANGED	Both still present; which is live is still unclear from the file layout alone.

7. Updated Recommended Next Steps

1. **Confirm the `/config/s3` debug endpoint is acceptable behind auth-only** — it's no longer publicly reachable (fix #1 covers it), but it's still not admin-gated specifically; anyone with a valid session token, not just admins, can read AWS config state. Decide if that's the intended blast radius.
2. **Add a baseline automated test suite + CI** — still nothing here; now more overdue given a second feature wave (phased workflow) has shipped with zero regression coverage.
3. **Delete the confirmed-dead code** — `s3.service.ts`, the four orphaned frontend AI files, the empty `project-ai-service.ts`, and one of the two signup forms. None of this was touched in the 26 Jul – 3 Aug push despite it being flagged in v1.0.
4. **Decide the fate of the decorative Rules engine** — unchanged; still only 2 of N rule types execute.
5. **Reconcile the two terminal implementations** — unchanged.
6. **Give `/product-modeling` a real backend model** — the mock-label fix is good interim honesty, but the underlying feature still has no persistence layer.
7. **Update CLAUDE.md / internal docs** — now more urgent: it undercounted routes by ~20 in v1.0, and the gap has grown by 9 more with the undocumented `/phases/*` group, plus the AI model version claim (gpt-4 vs. actual gpt-4o/gpt-4o-mini) is still wrong.

8. Consolidate the AiService rewrite commits — four same-week commits with near-identical messages ("implement AiService with OpenAI agentic tool support/function calling...") suggest either squash-worthy iteration or, worse, an actual merge/duplication problem worth a quick diff-review before it's forgotten.